

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Mathematics

Branch:- CSE/IT/ECE/MTC/PHY

Semester-I

Section: 1-8

Mid-Term Examination – October 2025

Subject Name / Code: Eng. Math-I, CSE/IT/ECE/MTC/PHY-1001

Max. Marks: 20
Time Allotted: 90 min

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

S. No.	Question	Marks
1.	(a). Let $f(x) = \sin x^2$ and $x = 0$. Find the <u>quadratic approximation</u> of $f(x)$, estimate $f(0.1)$, and provide an upper bound on the error using the third derivative.	3
	(b). Let $f(x)$ be differentiable on $\mathbb{R}$ and satisfy $ f'(x) \leq 2$ for all x . If $f(0) = 1$, find the maximum possible value of $f(5)$.	2
2.	(a). Examine the equality of $f_{xy}(0,0)$ and $f_{yx}(0,0)$ for the function $f(x,y) = \begin{cases} (x^2 + y^2) \tan^{-1}\left(\frac{y}{x}\right), & \text{if } x \neq 0, \\ \frac{\pi y^2}{2}, & \text{if } x = 0. \end{cases}$	3
	(b). Let $f(x,y) = x^2y + e^{xy}$, $\mathbf{r}(t) = (x(t), y(t)) = (t^2, \sin t)$, $t \in \mathbb{R}$. Compute $\frac{d}{dt}f(\mathbf{r}(t))$ using the chain rule. Express your answer in terms of the gradient ∇f and the derivative of the vector function $\mathbf{r}'(t)$.	2

S. No.	Question	Marks
3.	(a). Let the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$ (i) Find the directional derivative of f at the point $(0, 0)$ in the direction of the vector $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$. (ii) Examine whether the function f is differentiable at $(0, 0)$. If possible, determine the equation of the tangent plane to the surface $z = f(x, y)$ at the origin.	3
	(b). Examine the convergence of the series $\sum_{n=1}^{\infty} \frac{(2x - 3)^n}{n^2 5^n}.$ Also, find the interval of convergence.	2
4.	(a). Consider the function $f(x, y) = x^2 + y^4$ subject to the constraint $g(x, y) = y^2 - x = 0.$ Analyze the behavior of the function at the stationary point (using Bordered Hessian) along the constraint to determine whether it is a local maximum, minimum, or saddle point.	3
	(b). Let $f(x, y) = x^2 + y^2 - xy - x^3.$ Find all the stationary points of f and determine their nature using the Hessian test.	2

****ALL THE BEST****